RE 3: CSTR study: The reaction of t-butyl chloride and water

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Group B 21
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1 Introduction

In this experiment, the reaction between tert-butyl chloride and water under varying conditions was investigated. Two prepared solutions, consisting of 1.36 mL tert-butyl chloride in 998.64 mL methanol and 315 mL H_2O in 685 mL methanol, were pumped into a CSTR reactor with pump levels alternating between 60 and 80. A pH meter was used to monitor the conversion. A total of six runs were performed with varying temperatures and pump levels, where the pH level was measured every two minutes, five times during each run. The noted temperature, pH-level and pump level was used to determine the tBC conversion, activation energy and forward rate constant.

2 Theory

2.1 How does the reaction work

When water and t-bytyl choride reacts they can follow these two equations

$$(CH_3)_3CCl + H_2O- > (CH_3)_3COH + HCl \tag{2.1}$$

$$(CH_3)_3CCl + H_2O- > (CH_3)_2CCH_2 + H_3O^+Cl^- \tag{2.2}$$

The first reaction, 2.1, is a substitution reaction, and the second, 2.2, is an elimination reaction. Both of these happens in the system and in competition of each other. The slow step both of these reactions is the generation of the carbocation making this a first order reaction. Because of this the rate expression can be written as

$$-r = kC_{t-BC}. \tag{2.3}$$

From this equation the k is the reaction rate constant that can be found from the Arrheinus equation

$$k = Ae^{-E/RT}, \tag{2.4}$$

where A is the pre-exponetial factor, E is the activation energy, R is the gass constant and T is the temperature in kelvin.

2.2 How will the measured values be used to calculate kinetic parameters

The mass balance in a in a CSTR-system can be written as

$$\nu_f(C_i) - \nu_p(C_i)_p = -rV \tag{2.5}$$

Where ν is the volumetric flow rate of feed, and V is the reaction volume. When steady-state comnditions can be assumed ($\nu_f = \nu_p = \nu_0$) the mass balance equation can be rewritten to

$$(C_i)_f - (C_i)_p = -r\frac{V}{\nu_0} = -r\tau. \tag{2.6}$$

Where τ is the mean residence time. When this expression is inserted in the expression for a first order reaction 2.3 it will look like

$$(C_i)_f - (C_i)_p = k(C_i)_p\tau, \tag{2.7}$$

which, when solved for the rate constant, can look like

$$k = \frac{1}{\tau}[\frac{(C_i)_f}{(C_I)p} - 1] \tag{2.8}$$

From the Arrheinus equation 2.4 the activation energy can be calculated

$$ln(k) = ln(A) - \frac{E}{R}(\frac{1}{T}) \tag{2.9}$$

2.3 Plotting

When plotting equation 2.9 against $\frac{1}{T}$ it will give a linear graph with a slope of $\frac{-E}{R}$, this is where the activation energy is calculated. The pre-exponential factor for the Arrheinus equation can also be found from the plot.

3 Experimental

The experiment will be run at six different sets of operating conditions. The operation conditions are given in the table 3.1.

Table 3.1: Operation conditions

Run	Water bath temp.	Pump level
1	35	80
2	35	60
3	40	80
4	40	60
5	45	80
6	45	60

For the experiment, one solution of $17.5M$ H_2O in methanol, and one solution of $0.0125M$ t-butyl chloride in methanol is prepared. To start the experiment, first the water bath temperature is set. Then the feed-hoses are connected to the reactant solutions. The magnetic stirrer is turned on, and the pH-meter is re-calibrated using the available buffer solutions. The pump-level is set. The pH-value, the elapsed time, and the reactor's temperature is recorded in the lab journal every two minutes for 10 minutes in total for each of the six runs, as the reactor is approaching steady-state, at each experimental condition.

4 Results

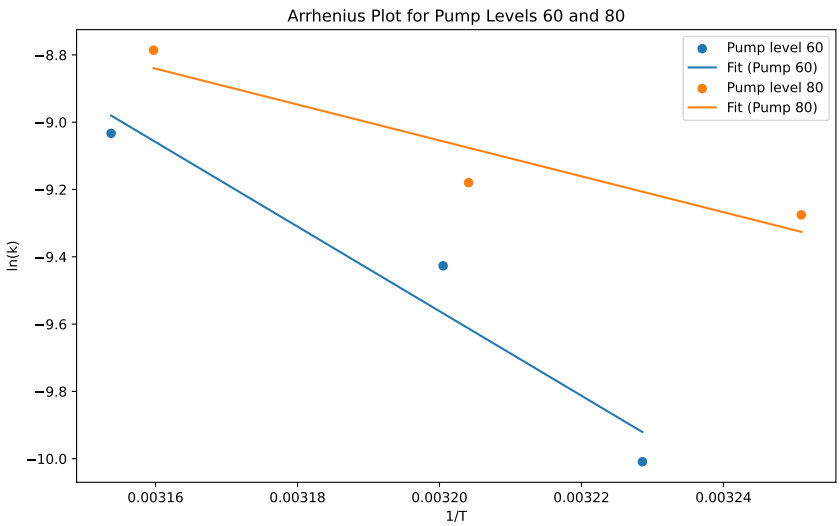

Figure 4.1: Linear regression plot

From the plot the activation energy and the pre-exponential factor from the Arrheinus equation can be calculated with the equations B.5 and B.6 They are represented in the table below

Table 4.1: The calculated activation energy and pre-exponential factor for both pump levels

Pump level	Activation energy, E_a	Pre-exponential factor, A
60	104.50	0.01485
80	0.67	$9.064 \cdot 10^{-4}$

During the experiment the pH-levels of the product flow was measured. The mean values for pH and temperature from the different runs is shown in table 4.2.

Table 4.2: Average pH levels and temperature for the six different runs

Run	Pump level	mean $T_{Reactor}$[°C]	mean pH
1	60	36.73	3.14
2	80	34.60	2.95
3	80	39.10	2.92
4	60	39.45	2.91
5	80	43.48	2.78
6	60	44.08	2.76

5 Discussion

The activation energy for pump levels 60 and 80 was measured to 104.5kJ and 0.67kJ respectively. These results are unusual, since the reactions are identical one would expect that the activation energy for each of the pump levels should be similar. The pre-exponential factor A was determined to be unusually low as well. A cause for this may be that since the reactants' concentrations were decreased by half, the possibility for collision decreases. A was found to be larger for pump level 60. This was expected, as the longer residence time should increase product yield. The table C.2 shows that there was a small amount of reactants used. This may be a cause for low pre-exponential factor values.

Generally, the pH-level decreases somewhat when the pump level decreases. This is expected, and indicates an increase in H^+-ions, which are a product in both the substitution and the elimination reaction. The equations for the reactions are given in 2.1 and 2.2 respectively. This can also be explained mathematically by the equations2.6 and 2.8. The reaction rate constant, k, is dependent on the residence time, τ. As shown in the equation 2.8, a high residence time will result in a lower reaction rate over time, leading to a decreased amount of product per time. The total amount of product accumulated in the residence time interval will still yield a higher product percentage, as reactants will still be present in the reactor.

For the first run of the experiment, table A.1 shows that the temperature fluctuated between 34.3℃ and 40.0℃. This doesn't appear to affect the pH-levels much, as the table shows the pH to decrease over time, as expected.

For the runs 2 to 6, shown in the tables A.2 through A.6, it was observed that there were relative small diffenrence in the pH measured. This is unexpected, as mathematically, the pH should decrease with higher temperatures, as is shown in 2.9.

After the final run, it appeared that a larger volume of one of the solutions—either t-butyl chloride or water—had been pumped into the reactor, even though both solutions were supposed to be pumped at the same rate. This was not confirmed, and the remaining volumes in the flasks were not measured. A reduced flow of t-butyl chloride would lead to a lower reaction rate, as the already low concentration of t-butyl chloride would decrease even further. If this were the case, a higher flow rate of the water solution would likely occur, potentially halting the pH decrease or even causing an increase, since the water solution is more basic than the reactor solution.

6 Conclusion

The experiment showed that pH levels generally decreased somewhat as pump levels decreased, indicating an increase in H^+ ions. Longer residence times, resulting from lower pump levels, led to a lower reaction rate over time but still yielded a higher overall percentage of product. Higher temperatures also lowered pH by increasing the reaction rate constant, producing more product. In the first run, temperature fluctuations had little effect on pH, while runs 2-6 showed unexpectedly small pH differences. Activation energies for pump levels 60 and 80 differed significantly, which was unusual since the reactions were identical. The pre-exponential factor A was also unexpectedly low, likely due to decreased reactant concentrations. A possible imbalance in the volumes of the t-butyl chloride mixture and water in methanol pumped into the reactor may have affected the reaction rate and pH. As this is merely a potential source of error, no definitive conclusions can be drawn from this.

A Experimental data

Table A.1: Run 1, flow rate at 60mL s^{-1}

Time	Temperature	pH
0	34.3	3.22
2	34.6	3.18
4	34.7	3.16
6	40	3.13
8	38.2	3.09
10	38.6	3.06

Table A.2: Run 2, flow rate at 80mL s^{-1}

Time	Temperature	pH
0	34.4	2.91
2	34.7	2.93
4	34.7	2.94
6	34.7	2.96
8	34.6	2.97
10	34.5	2.98

Table A.3: Run 3, flow rate at 80mL s^{-1}

Time	Temperature	pH
0	39.3	2.92
2	38.9	2.92
4	39.0	2.93
6	39.2	2.92
8	39.1	2.93
10	39.1	2.91

Table A.4: Run 4, flow rate at 60mL s^{-1}

Time	Temperature	pH
0	39.3	2.93
2	39.4	2.92
4	39.5	2.92
6	39.5	2.91
8	39.5	2.90
10	39.5	2.90

Table A.5: Run 5, flow rate at 80mL s^{-1}

Time	Temperature	pH
0	44.2	2.78
2	43.4	2.77
4	43.2	2.78
6	43.1	2.79
8	43.5	2.79
10	43.5	2.79

Table A.6: Run 6, flow rate at 60milliL/s

Time	Temperature	pH
0	43.1	2.79
2	44.0	2.77
4	44.2	2.76
6	44.4	2.75
8	44.4	2.74
10	44.4	2.74

B Calculations

B.1 Volumetric flow rate and residence time

The calibration curve used to convert pump level to flow rate was given as

$$Q = 0.0022x + 0.025, \tag{B.1}$$

Where the x is the pump level and Q is flow rate. To get the residence time the volume of the reactor is divided by the flow rate

$$\tau = \frac{V}{Q} \tag{B.2}$$

the table B.1 shows the calculated values

Table B.1: The calculated volumetric flow rate, and residence time of the mixture in the reaction tank for each of the pump levels.

Pump level	Volumetric flow rate [mL s^{-1}]	Residence time [s]
60	0.157	1318
80	0.201	1030

B.2 Concentration and conversion

The output concentration of t-butyl chloride can be calculated by using the pH to determine the proton concentration and subtracting this from the initial t-butyl chloride concentration. The number of protons contributed by the water is equal to the amount of t-butyl chloride that has reacted. This is shown in Eq. (B.3).

$$(C_{(CH_3)_3CCl})_{out} = (C_{(CH_3)_3CCl})_{in} - 10^{-pH} \tag{B.3}$$

The conversion rate is calculated using Eq. (B.4).

$$X = \frac{(C_i)_{in} - (C_i)_{out}}{(C_i)_{in}} \tag{B.4}$$

Table B.2 presents the calculated output concentration of t-butyl chloride and the calculated conversion rates for each run.

Table B.2: Calculated X values for each run

Run	$(C_{(CH_3)_3CCl})_{out}$	X
1	0.0118	0.056
2	0.0114	0.088
3	0.0013	0.896
4	0.0013	0.896
5	0.0108	0.136
6	0.0108	0.136

B.3 Reaction rate and reaction constant

In order to calculate the reaction rate for each run the equation 2.6 with respect to r wa used, and for the reaction constant the equation 2.8 was used. Resulting in the values in the table B.3.

Table B.3: Reaction rate and reaction constant for every run

Run	-r [$\mathrm{mol\,L^{-1}\,s}$]	k [$\mathrm{mol\,s^{-1}}$]
1	$5.309 \cdot 10^{-5}$	$4.499 \cdot 10^{-5}$
2	$1,068 \cdot 10^{-6}$	$9.369 \cdot 10^{-5}$
3	$1.165 \cdot 10^{-6}$	$1.031 \cdot 10^{-4}$
4	$9.101 \cdot 10^{-7}$	$8.054 \cdot 10^{-5}$
5	$1.651 \cdot 10^{-6}$	$1.528 \cdot 10^{-4}$
6	$1.289 \cdot 10^{-6}$	$1.194 \cdot 10^{-4}$

When plotting the linear regression for the Arrheinus equation as shown in figure 4.1 the slope and the intersection with the y-axis was found and presented in table **??**

Table B.4: Overview of the different slopes and intersections

Pump level	Slope	Intersection with y-axis
60	-210.464	0.01485
80	-80.325	$9.064 \cdot 10^{-4}$

The activation energy, E_a, can be found using the equation

$$E_a = -slope \cdot R, \tag{B.5}$$

where the slope is found using python, and R is the gas constant, and the pre-exponential factor, A, can be found using the equation

$$A = e^{intersectional\ value}. \tag{B.6}$$

Where the intersectional value is the value in the intersection with the y-axis

C Example calculations (Work plan)

The fluid flow can be calculated from the equation

$$y = 0.0022x + 0.025, \tag{C.1}$$

Where y is the fluid flow, and x is the pump level. When multiplied with the amount of seconds the pump was on and summed up, the total consumption of reactants can be found. The table under show a general overview of the consummation

Table C.1: General consumption of solution

Run	Pump level [mL s^{-1}]	Runtime [s]	Consumption [mL]	
			H2O	tBC
1	80	600	120.6	120.6
2	60	600	94.2	94.2
3	80	600	120.6	120.6
4	60	600	94.2	94.2
5	80	600	120.6	120.6
6	60	600	94.2	94.2
Sum			644.4	644.4

When the volumetric percentage from making the solutions is factored in, the volumetric amount of solvent and reactants used can be determined. They can be found in the table below

Table C.2: Consuption of reactants and solvents

tBC [mL]	H_2O [mL]	$\Sigma MeOH$ [mL]
8.8	203	1084